

Algorithmic Foundations of Consumer Food Scoring Engines: Translating Nutrient Profiling and Additive Toxicology into Quantitative Risk Metrics

Algorithmic Architecture and Nutrient Profiling Models

The deployment of consumer-facing food scoring engines requires a robust mathematical and toxicological framework capable of translating multi-dimensional nutritional data and chemical hazards into intuitive, actionable ratings. Standardized profiling schemes, such as the European Nutri-Score, represent the foundational baseline for these algorithms. Originally derived from the United Kingdom Food Standards Agency (FSA) and Office of Communications (Ofcom) nutrient profile model—known taxonomically as "model WXYfm"—the classic Nutri-Score algorithm converts mandatory back-of-pack nutritional facts per 100 grams or milliliters into a continuous ordinal scale graded from A (highest nutritional quality) to E (lowest nutritional quality). The underlying mathematical mechanism of model WXYfm allocates discrete points based on nutrient density. Negative points (\$0\$ to \$10\$) are assigned for energy, saturated fatty acids, total sugars, and sodium. These are offset by positive points (\$0\$ to \$5\$ or \$0\$ to \$7\$) allocated for dietary fiber, proteins, and the percentage of fruits, vegetables, legumes, and nuts. The baseline allocation for negative points follows strict concentration thresholds:

Points	Energy (kJ)	Saturated Fat (g)	Total Sugar (g)	Sodium (mg)
0	$0 \leq 335$	$0 \leq 1.0$	$0 \leq 4.5$	$0 \leq 90$
1	$335 < 1.0 \leq 4.5$	$1.0 < 2.0 \leq 4.5$	$4.5 < 9.0 \leq 18.0$	$90 < 100 \leq 135$
2	$4.5 < 9.0 \leq 18.0$	$2.0 < 3.0 \leq 4.5$	$9.0 < 18.0 \leq 27.0$	$100 < 135 \leq 167$
3	$18.0 < 27.0 \leq 36.0$	$3.0 < 4.0 \leq 5.0$	$27.0 < 36.0 \leq 45.0$	$135 < 167 \leq 201$
4	$27.0 < 36.0 \leq 45.0$	$4.0 < 5.0 \leq 6.0$	$45.0 < 54.0 \leq 63.0$	$167 < 201 \leq 268$
5	$36.0 < 45.0 \leq 54.0$	$5.0 < 6.0 \leq 7.0$	$63.0 < 72.0 \leq 81.0$	$201 < 268 \leq 335$
6	$45.0 < 54.0 \leq 63.0$	$6.0 < 7.0 \leq 8.0$	$81.0 < 90.0 \leq 100.0$	$268 < 335 \leq 450$
7	$54.0 < 63.0 \leq 72.0$	$7.0 < 8.0 \leq 9.0$	$90.0 < 100.0 \leq 110.0$	$335 < 450 \leq 900$

While the Nutri-Score—championed by Professor Hercberg's research team and supported by the International Agency for Research on Cancer (IARC)—succeeds in classifying overall nutritional quality, it exhibits major structural limitations. Most notably, it fails to evaluate the presence of chemical additives, pesticide residues, or the degree of industrial food processing. Consequently, advanced consumer engines have developed hybrid multi-criteria scoring models. The Yuka platform, for instance, utilizes a composite scoring architecture. Within its 100-point scale, 60% of the weight is assigned to the nutritional profile (calculated using the original, non-updated Nutri-Score algorithm to prevent threshold inconsistencies). An additional 10% bonus is allocated for certified organic status, which serves as an administrative proxy for the absence of synthetic chemical pesticides. The remaining 30% of the score is determined by an independent food additive risk-level classification system. Similarly, the Environmental Working Group (EWG) Food Scores engine rates products on a 1-to-10 scale (1 representing optimal quality and 10 representing maximum hazard). The EWG algorithm differentiates itself by anchoring the overall rating on nutritional quality, then adding sequential penalties based on "ingredient concerns" (including contaminants, pesticides, hormones, and additives) and "processing scores" (which measure physical and chemical modifications of whole

ingredients). Scoring Engine Output Metrics Primary Profiling System Additive Penalty
 Logic Organic / Pesticide Integration Primary Structural Vulnerabilities Nutri-Score Five-
 Class Grade (A to E) Modified UK FSA Model WXYfm Omitted from calculation No direct
 integration Susceptible to formulation manipulation; ignores chemical load. Yuka
 Continuous Scale (\$0\text{ to }100\$) Original Nutri-Score (\$60\%\$ weight) Stepwise
 subtraction based on four hazard tiers (\$30\%\$ weight) \$10\%\$ flat bonus for certified
 organic status Treats ingredients in isolation; ignores concentration-dependent risk
 curves. EWG Food Scores Inverse Scale (\$1\text{ to }10\$) Modified Oxford/FSA Profile
 (Major Anchor) Baseline substance hazard adjusted by food exposure weights \$+0.5\$
 penalty for non-organic; pesticide risk scaled by produce lists Mathematical complexity
 introduces linear penalty scaling errors. Siga Technological Grade (\$1\text{ to }100\$) Multi-
 tier Matrix and Processing Evaluation Identified as Markers of Ultra-Processing (MUPs)
 Focused on processing intensity; organic status secondary High data-entry requirements;
 complex regulatory verification. Mathematical Modeling of Additive and Contaminant
 Penalties To construct an academically defensible scoring engine, developers must
 formalize the mathematical relationships that govern additive and contaminant penalties,
 avoiding the scientific distortions associated with simplistic binary hazard lists. Linear
 additive scoring systems, while computationally inexpensive, often fail to model the non-
 linear toxicological realities of dose-response relationships and cumulative exposure. The
 Yuka additive scoring engine utilizes a flat-subtraction hazard-tier model. Each identified
 food additive is mapped to one of four hazard tiers, which trigger direct subtractions from
 the 100-point baseline : No Risk (Green Dot): Deducts \$0\$ points. Limited Risk (Yellow Dot):
 Deducts \$6\$ points. Moderate Risk (Orange Dot): Deducts \$15\$ points. High Risk /
 Hazardous (Red Dot): Deducts \$30\$ points, while concurrently triggering a hard
 mathematical cap that limits the maximum possible score of the finished product to
 \$49/100\$. While this heuristic ensures that any product containing a high-risk additive is
 classified as "Mediocre" or "Poor," it does not account for the concentration of the additive
 or its interaction with the food matrix. The EWG Food Scores engine addresses this by
 implementing a dynamic, non-linear penalty calculation. The overall EWG score is derived
 using the following core equation :
$$\text{Overall Score} = S_{\text{nutrition}} + \left(S_{\text{ingredient_concern}} \times W_{\text{ingredient}} \right) + \left(S_{\text{processing}} \times W_{\text{processing}} \right) + \Delta_{\text{conventional}}$$
 Where $S_{\text{nutrition}}$ is the baseline nutrition score anchoring the product. $S_{\text{ingredient_concern}}$ represents the aggregate chemical
 and biological hazard score on a scale of \$1\$ to \$10\$. $W_{\text{ingredient}}$ is a dynamic
 weighting factor configured to scale non-linearly with the severity of the hazard. For low-
 hazard ingredients ($S_{\text{ingredient_concern}} < 5$), $W_{\text{ingredient}} = 0.2$. If
 any ingredient concern exceeds a score of \$5\$, the weight scales linearly to a maximum of

\$0.4\$ when $S_{\text{ingredient_concern}} = 10$, adding up to \$4.0\$ penalty points to the overall score. The ingredient concern score itself is modeled as a function of the chemical's intrinsic toxicological hazard and food-specific exposure factors

$$S_{\text{ingredient_concern}} = S_{\text{substance}} \times$$

$F_{\text{food_weighting}}$ Here, $S_{\text{substance}}$ is the raw hazard score of the substance, reflecting its independent carcinogenic, reproductive, or immunological toxicity. The food-weighting factor ($F_{\text{food_weighting}}$) adjusts this value based on empirical dietary exposure data. For example, the food-weighting factors for methylmercury in seafood are stratified by mean concentration

$$F_{\text{food_weighting}} =$$

$\begin{cases}$

$$1.0 \ \& \ \text{if } [\text{Hg}] > 0.25 \ \text{ppm (Very High Concern)} \ \backslash \backslash$$

$$0.6 \ \& \ \text{if } 0.1 \ \text{ppm} < [\text{Hg}] \leq 0.25 \ \text{ppm (Moderate Concern)}$$

$\end{cases}$ This ensures that high-exposure seafood like swordfish steaks is heavily penalized, whereas low-exposure species remain unpenalized. For pesticide residues on conventionally grown agricultural products, EWG adjusts the weight based on its annual monitoring databases. Produce listed on the "Dirty Dozen" receives an

$F_{\text{food_weighting}} = 0.3$, products on the "Clean Fifteen" receive \$0.2\$, and unlisted conventional products receive \$0.25\$, while certified organic produce defaults to \$0\$. The physical and chemical processing of ingredients is modeled through EWG's nested processing score equations. For each individual ingredient, a processing score is

$$S_{\text{ingredient}} = 0.2 + w_{\text{source}} + \sum$$

$w_{\text{processing_weight}}$ Where $w_{\text{source}} = 1.0$ for petroleum-derived synthetic substances and \$0\$ for natural matrices. The term $\sum$

$w_{\text{processing_weight}}$ aggregates the typical industrial steps (e.g., chemical extraction, hydrogenation, polymerization) required to manufacture the ingredient. Highly refined ingredients are assigned specialized baseline processing scores: non-artificial added sweeteners receive \$3.0\$, chemically extracted vitamins and minerals receive \$3.0\$, spices receive \$0.7\$, and herbs receive \$0.5\$. Synthetic industrial chemicals and generic "flavors" bypass this formula and are assigned a maximum baseline of \$5.0\$. To

aggregate these values into a single product-level metric, the algorithm applies an exponential decay weighting scheme to prioritize the primary ingredients

$$\text{Combined Ingredient Score} = 5 \times S_{\text{first_ingredient}} + 3 \times$$

$$S_{\text{second_ingredient}} + \sum_{i=3}^n S_{\text{ingredient}_i}$$

The final processing score is normalized to a 10-point scale: $\text{Overall Processing Score} = 5 \times$

$$S_{\text{product_level}} + \text{Combined Ingredient Score}$$

$$\text{Final Processing Score}$$

$$\text{Score} = \frac{\text{Overall Processing Score} - 1}{10}$$
 When the Final Processing Score exceeds 5\$, its corresponding weighting factor ($W_{\text{processing}}$) in the overall score formula scales from 0.1\$ to 0.2\$, contributing a maximum of 2.0\$ points to the overall product-level penalty. This sophisticated mathematical structure represents a robust attempt to quantify processing intensity, though it remains constrained by its reliance on linear scaling parameters.

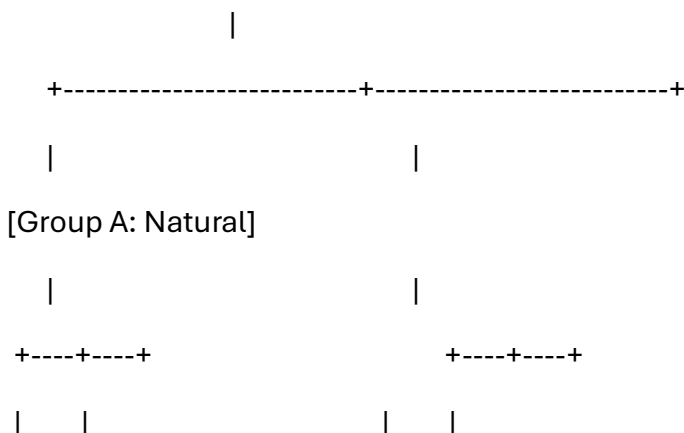
Matrix Effects and the Food Processing Spectrum: From NOVA to Siga

A major criticism of first-generation scoring models is their failure to account for food matrix structures and the systemic health impacts of industrial processing. Ultra-processed foods (UPFs) are typically characterized not merely by poor macronutrient ratios, but by the complete physical deconstruction and chemical reconstruction of the food matrix. This structural alteration accelerates ingestion rates, alters glycemic responses, and disrupts natural satiety signaling pathways.

The dominant paradigm for processing classification is the NOVA system, developed by Professor Monteiro's team. NOVA categorizes all foods into four mutually exclusive groups based on the nature and purpose of processing:

- Group 1: Unprocessed or minimally processed foods (e.g., fresh vegetables, seeds, muscle meats, plain yogurt).
- Group 2: Processed culinary ingredients (e.g., vegetable oils, butter, sugar, salt derived from Group 1 foods by physical pressing or milling).
- Group 3: Processed foods (e.g., canned vegetables, simple cheeses, freshly made artisanal breads combining Group 1 and Group 2 ingredients).
- Group 4: Ultra-processed foods (UPFs), defined as industrial formulations consisting of multiple ingredients and additives designed to make the final product highly palatable, shelf-stable, and hyperpalatable.

While NOVA successfully identifies ultra-processed formulations, it lacks the technical granularity required by food scientists and retailers to optimize product formulations. This limitation led to the development of the Siga classification system in 2020 by a French start-up. Siga refines the four holistic NOVA categories into nine distinct technological subgroups. This system evaluates three core parameters: the physical impact of processing on the food/ingredient matrix, the nutritional density of added fats, sugars, and salts, and the nature and number of Markers of Ultra-Processing (MUPs).



[A0] [A1]

(Intact) (Degraded)

(Balanced) (High)

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[Group C: Ultra-Processed]

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[C1 to C3]

(Purified/Denatured/MUPs)

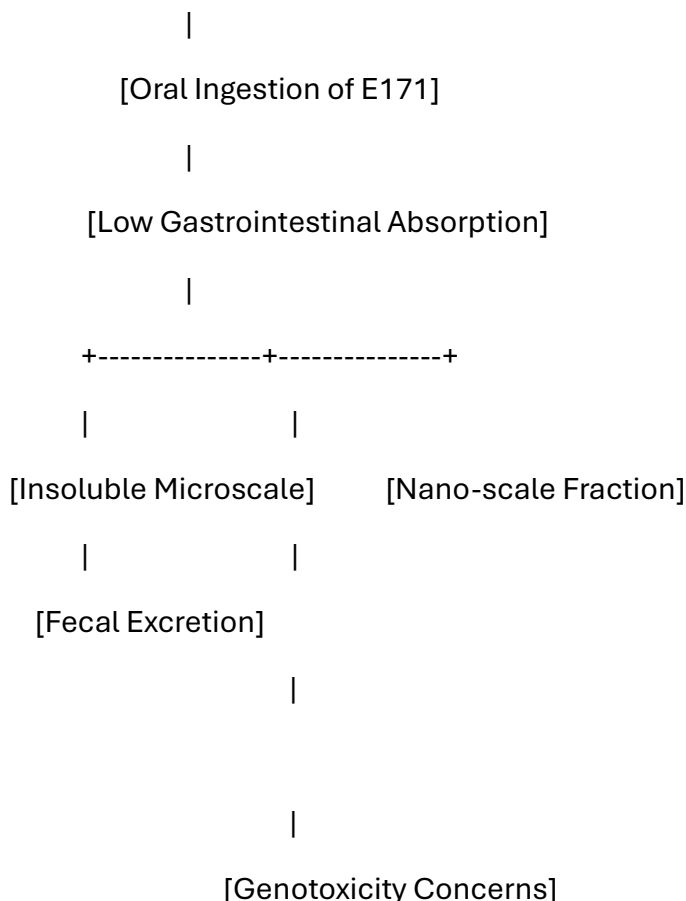
Siga categorizes foods based on specific matrix and ingredient criteria :Group A (Minimally Processed): Subdivided into Group A0 (intact initial matrix like fresh fruits or pasteurized milk) and Group A1 (degraded raw matrix like fresh fruit juices, ground meat, or unrefined culinary ingredients).Group B (Processed): Products combining Group A foods with culinary ingredients. Subgroup B1 contains added sugars, salts, and fats within healthy dietary guidelines, whereas Subgroup B2 contains at least one of these nutrients exceeding recommended healthy thresholds.Group C (Ultra-Processed): Defined by the presence of at least one purified, denatured, or synthetic ingredient. This group is further stratified from C1 to C3 based on the number and toxicological risk of at-risk additives and MUPs.Siga's criteria for identifying Markers of Ultra-Processing rely heavily on the functional classes of food additives defined by the United Nations Codex Alimentarius. Any additive classified as "cosmetic"—meaning its primary technological function is to restore, mimic, or enhance sensory properties (taste, aroma, color, texture) that were lost during industrial processing—is classified as an MUP. These functional classes include anti-foaming agents, foaming agents, bulking agents, gelling agents, thickeners, carbonating agents, colors, emulsifiers, emulsifying salts, flavor enhancers, glazing agents, humectants, and non-sugar sweeteners.A systematic literature review conducted in December 2023 following PRISMA guidelines evaluated six highly processed food classification systems: NOVA, EPIC, IFPRI, UNC, UP3, and Siga. The review concluded that while most systems lack functional specificity, Siga stands out by addressing the diverse risk profiles of individual additives and incorporating quantitative nutritional quality criteria alongside processing indicators. This multi-dimensional approach aligns with epidemiological data indicating that the health risks of ultra-processed foods are highly heterogeneous.For example, large-scale cohort studies demonstrate that the increased risk of cardiovascular disease, type 2 diabetes, and obesity associated with total UPF intake is heavily driven by specific subcategories, notably processed meats, sugar-sweetened beverages, and ultra-processed sauces. Conversely, ultra-processed wholegrain breads, yogurts, and dairy-based desserts are frequently associated with a neutral or reduced risk of cardiometabolic

disease. This underscores the critical need for scoring engines to transition from blanket UPF penalties to highly granular, substance-specific risk models.

The Titanium Dioxide (E171) Regulatory Schism: Nanotoxicology and Exposure Realities

The regulatory history of Titanium Dioxide (E171) highlights a profound scientific division among international food safety authorities regarding the translation of nanotoxicological hazard data into human exposure risks. Historically approved globally as a white food colorant, E171 consists of a polydisperse mixture of titanium dioxide (TiO_2) particles. While pigment-grade E171 is primarily composed of microscale particles, up to 50% of the particles by number fall within the nanoscale range ($< 100\text{ nm}$). This nanoscale fraction became the focus of intense scientific scrutiny. At the nano-scale, physical and chemical properties change dramatically; particles exhibit increased surface area, enhanced reactivity, and the potential to cross biological membranes, accumulate in organs, and interact directly with cellular structures.

[E171 Ingestion Pathway]



In May 2021, the European Food Safety Authority (EFSA) published an updated safety assessment of E171. Led by the Panel on Food Additives and Flavourings (FAF), chaired by Professor Maged Younes, and supported by a specialized working group chaired by Professor Matthew Wright, EFSA applied its 2018 Nanotechnology Guidance to the

evaluation for the first time. The FAF Panel concluded that while gastrointestinal absorption of ingested TiO_2 is low ($< 0.1\%$), the absorbed nanoscale fraction possesses an extremely long biological half-life, leading to systemic accumulation in organs such as the liver, spleen, and mesenteric lymph nodes. Crucially, the panel concluded that a concern for genotoxicity could not be ruled out. Genotoxicity refers to the ability of a chemical agent to damage cellular DNA, potentially initiating carcinogenic cascades. EFSA's review of in vitro and in vivo assays indicated that TiO_2 nanoparticles have the potential to induce DNA strand breaks and chromosomal damage, though they do not appear to cause direct gene mutations. Because several modes of action (such as the generation of reactive oxygen species and physical interference with mitotic spindles) could operate in parallel without a clear dose-response threshold, EFSA concluded it was impossible to establish a safe Acceptable Daily Intake (ADI). This hazard-based assessment led to a complete ban on E171 as a food additive across the European Union, enacted by SCoPAFF on October 8, 2021, and fully implemented on August 8, 2022, following years of advocacy by NGOs like SAFE. In sharp contrast, multiple international regulatory bodies, including the US Food and Drug Administration (FDA), Health Canada, Food Standards Australia New Zealand (FSANZ), and the UK Committee on Toxicity (COT), fundamentally disagreed with EFSA's conclusions. These agencies argued that EFSA made a manifest scientific error by failing to distinguish between highly dispersed, sonicated nanomaterials and real-world food-grade E171. In food matrices, E171 particles naturally agglomerate, and there is no physiological evidence that E171 de-agglomerates into single nanoparticles within the human gastrointestinal tract. Toxicological studies utilizing artificially sonicated nanoparticles suspended in liquid dispersion are not representative of human dietary exposure, as the natural food matrix prevents direct nano-bio interactions. Furthermore, dissenting regulators cited robust, direct experimental evidence:

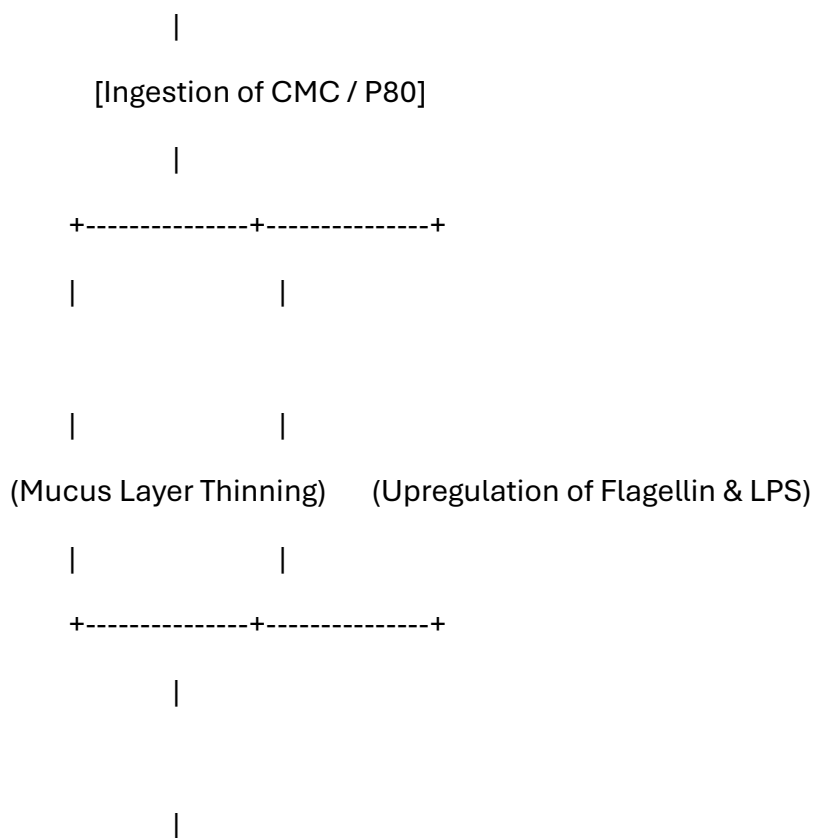
- The Extended One-Generation Reproductive Toxicity (EOGRT) Study: A gold-standard dietary study in rats exposed to food-grade E171 at doses up to 1000 mg/kg bw/day (homogeneously mixed into dry feed to preserve particle morphology) demonstrated no systemic, developmental, reproductive, or immunotoxicological adverse effects.
- The US National Cancer Institute (NCI) Bioassay: A landmark two-year carcinogenicity study in rats and mice showed no evidence of systemic toxicity or tumor induction at dietary concentrations of food-grade TiO_2 up to $50,000\text{ ppm}$ (equivalent to several thousand milligrams per kilogram of body weight per day).
- Human Absorption Profiles: Human clinical data confirmed that systemic circulation of TiO_2 after ingestion is negligible, and no toxicological hazards are observable under realistic exposure scenarios.

These regulatory positions are summarized below:

Jurisdiction / Agency	Current Authorization Status	Regulatory Limits	Primary Scientific Evidence Cited	Key Criticisms of EFSA's Position
European Union (EFSA)	Completely Banned (August 2022)	0%	2018	

Nanotechnology Guidance; evidence of tissue accumulation; in vitro DNA strand break assays. Precautionary hazard-based approach; did not identify a safe exposure threshold. United States (FDA) Authorized ≤ 1.0% by weight US NCI 2-year rodent bioassay; comprehensive exposure-risk models. Rejected genotoxicity claims; argued EFSA relied on sonicated nanoparticles not representative of diet. Canada (Health Canada) Authorized Good Manufacturing Practice (GMP) Comprehensive review of all post-EFSA toxicological databases. Found no evidence of immunotoxicity or reproductive harm; cited food matrix protection. Australia & New Zealand (FSANZ) Authorized Permitted in wide food categories Literature call-for-data; focus on intestinal agglomeration dynamics. Emphasized lack of in vivo genotoxicity from dietary administration of food-grade E171. United Kingdom (COT / FSA) Authorized / Under Review Permitted at approved levels Interim Position Paper (January 2022); review of the EOGRT study. Concluded weight of evidence does not support EFSA's genotoxicity claims. Xenobiotic Disruptors of Intestinal Homeostasis: Emulsifier-Induced Microbiome Dysbiosis While traditional regulatory toxicology focusing on mutagenicity and acute organ toxicity continues to shape public policy, recent clinical and metagenomic research demonstrates that several widely approved food additives cause chronic, low-grade inflammatory diseases by disrupting the intestinal microbiota and host epithelial barrier. The human gut is protected from its dense, complex microbiota by a stratified mucosal barrier. This barrier consists of a sterile inner mucus layer composed of heavily glycosylated mucin proteins (principally Muc2) and a more diffuse outer mucus layer where commensal bacteria reside. Synthetic emulsifiers, such as Sodium Carboxymethylcellulose (CMC) and Polysorbate 80 (P80), possess detergent-like amphiphilic properties. In vivo animal models have shown that relatively low dietary concentrations of CMC and P80 (1.0% in drinking water) induce low-grade intestinal inflammation, increase body fat pad weight, elevate fasting blood glucose concentrations, and promote robust colitis in genetically susceptible hosts. This pathology is accompanied by "microbiota encroachment"—the physical migration of intestinal bacteria across the mucin layer, reducing the distance between the microbiota and the host intestinal epithelial cells to less than a safe threshold. Importantly, this effect is mediated directly through the intestinal microbiota. When CMC and P80 were administered to human fecal microbiotas maintained ex vivo in the Mucosal Simulator of the Human Intestinal Microbial Ecosystem (M-SHIME)—a host-free, multi-stage continuous culture system—both emulsifiers acted directly upon the bacterial populations. This exposure caused a rapid upregulation of bioactive flagellin, a potent pro-inflammatory flagellar protein that acts as an agonist for host Toll-like receptor 5 (TLR5). The CMC-induced flagellin surge occurred rapidly (1 day) and was driven by transcriptional changes in microbial gene expression. In contrast, the P80-induced flagellin increase occurred more slowly and was associated with profound taxonomic shifts. When these emulsifier-conditioned ex vivo microbiotas

were transplanted into germ-free recipient mice, the recipients developed low-grade mucosal inflammation, microbial encroachment, and metabolic syndrome, demonstrating that the altered microbiota was both necessary and sufficient to drive the disease phenotype. Conversely, when CMC and P80 were administered to mice colonized exclusively with the simple, pathobiont-free Altered Schaedler Flora (ASF), which contains only eight defined bacterial species, no mucosal thinning, inflammation, or metabolic syndrome occurred. This confirms that complex, multi-species microbial ecosystems are required to mediate emulsifier-induced pathology. Metagenomic analyses utilizing dynamic human gut simulators (such as the BFBL simulator) have further elucidated these taxonomic shifts. Using 16S rDNA amplicon-based metagenomic sequencing, researchers demonstrated that P80 and CMC alter the gut microbiome in a dose-dependent manner, significantly decreasing beneficial, barrier-enhancing taxa while increasing mucin-degrading and pro-inflammatory groups. Polysorbate 80 exposure significantly reduces the abundance of *Bacteroides dorei* and *Akkermansia muciniphila*, taxa critical for maintaining mucin synthesis and anti-inflammatory homeostasis. Carboxymethylcellulose exposure drives a corresponding increase in the abundance of mucolytic taxa, including *Ruminococcus torques* and *Hungatella*, which actively degrade the host mucin layer and diminish epithelial barrier integrity.



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[Chronic Mucosal Colitis]

This mucolytic degradation pattern is also produced by food-grade hydrocolloids, specifically Carrageenan (CGN). Carrageenan is a high-molecular-weight sulfated polysaccharide extracted from red seaweeds. Metagenomic and chromatographic studies have identified that *Bacteroides xylanisolvens* strain C3, a common human gut symbiont, actively degrades food-grade lambda-carrageenan into low-molecular-weight degraded carrageenans (dCGNs) within the gastrointestinal tract. These dCGNs are potent inflammatory agents. Using high-performance liquid chromatography (HPLC) and thin-layer chromatography (TLC), researchers demonstrated that these microbially generated dCGNs stimulate host macrophages, upregulating the production of nitric oxide (NO) and cyclooxygenase-2 (COX-2). This inflammatory cascade is mediated by the activation of host Nrf2 and NLRP3 inflammasome pathways, driving the expression of pro-inflammatory cytokines, including interleukin-1 beta ($\text{IL-1}\beta$), tumor necrosis factor alpha ($\text{TNF-}\alpha$), and interleukin-6 (IL-6). Furthermore, intact carrageenan directly contacts host intestinal epithelial cells, activating the pro-inflammatory nuclear factor kappa B ($\text{NF-}\kappa\text{B}$) pathway via TLR4 receptor binding. This dual mechanism—microbially mediated degradation and direct epithelial receptor activation—results in severe damage to the mucosal barrier, downregulates protective short-chain fatty acids (SCFAs), and is linked to the development and relapse of inflammatory bowel diseases, including ulcerative colitis and Crohn's disease.

Toxicological Re-evaluations and Behavioral Hazard Discrepancies

Constructing a robust consumer scoring engine requires careful calibration of behavioral hazards and chemical evaluations, particularly where major regulatory bodies disagree. The evaluation of the non-sugar sweetener Aspartame and synthetic azo food colorants illustrates these toxicological discrepancies. In July 2023, two independent World Health Organization (WHO) bodies released concurrent evaluations of Aspartame. The International Agency for Research on Cancer (IARC) Monographs Working Group, during its 134th meeting, classified aspartame as "possibly carcinogenic to humans" (Group 2B). This classification was based on "limited evidence" of hepatocellular carcinoma in humans, derived from three prospective cohort studies: the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort, a pooled analysis of the NIH-AARP Diet and Health Study and the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial cohorts, and the Cancer Prevention Study II (CPS-II) cohort. Artificially sweetened beverage consumption was utilized as an exposure proxy. IARC also identified "limited evidence" of carcinogenicity in experimental animals,

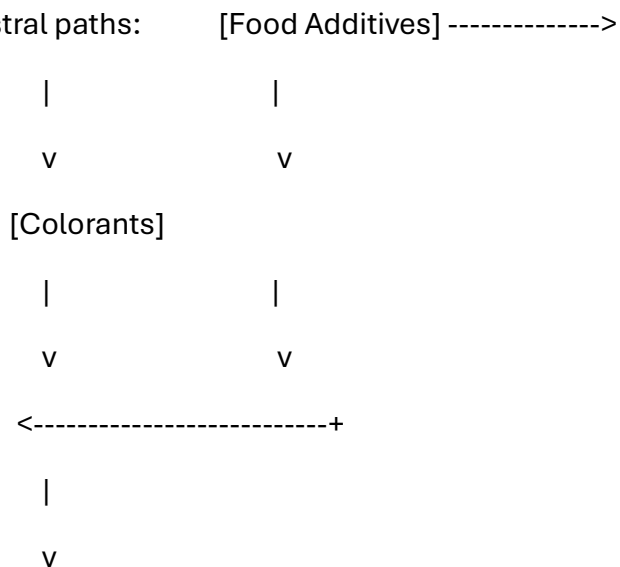
citing an increased incidence of malignant neoplasms across rodent studies, and "limited mechanistic evidence" showing that aspartame induces oxidative stress and chronic inflammation. Conversely, the Joint FAO/WHO Expert Committee on Food Additives (JECFA) conducted a comprehensive risk assessment and concluded there was no convincing evidence of adverse systemic or carcinogenic effects from ingested aspartame. JECFA reaffirmed the established Acceptable Daily Intake (ADI) of 40 mg/kg bw , a position supported by the US FDA, Health Canada, and EFSA. This risk-based conclusion is supported by aspartame's rapid metabolic degradation. Upon ingestion, aspartame is completely hydrolyzed within the gastrointestinal tract by intestinal esterases and peptidases into three common dietary metabolites: phenylalanine, aspartic acid, and methanol. Because these metabolites are identical to those absorbed in much larger quantities from common, unprocessed foods, no intact aspartame enters the systemic circulation. Human clinical studies at doses up to the ADI showed no increases in the plasma concentrations of these metabolites. JECFA and the FDA evaluated twelve oral carcinogenicity studies, identifying severe methodological limitations in the positive findings reported by Soffritti et al. (2005, 2006, 2007, 2010). These studies failed to adjust for litter effects in prenatal cohorts and suffered from diagnostic confounding due to chronic respiratory infections, which can yield false positives for lymphoma and lung lesions. Consequently, both JECFA and the FDA concluded that the epidemiological associations between aspartame and liver cancer were likely due to chance, selection bias, and lifestyle confounding, maintaining that the current ADI is protective of public health. A similar regulatory divergence occurred following the landmark "Southampton Study" on synthetic food colors and childhood behavior. Commissioned by the UK Food Standards Agency (FSA), McCann et al. (2007) conducted a randomized, double-blind, placebo-controlled food challenge with 153 3-year-olds and 144 8/9-year-olds from the general population. This study followed an earlier preschool trial by Bateman et al. (2004) on the Isle of Wight. The children were challenged daily over a six-week period with one of two active drink mixtures containing artificial food colors and the preservative Sodium Benzoate (E211), or an identical placebo fruit juice: Active Mix A: Tartrazine (E102), Sunset Yellow FCF (E110), Carmoisine (E122), Ponceau 4R (E124), and Sodium Benzoate (E211). Active Mix B: Quinoline Yellow (E104), Sunset Yellow FCF (E110), Allura Red AC (E129), Ponceau 4R (E124), and Sodium Benzoate (E211). Behavioral changes were quantified using a standardized Global Hyperactivity Aggregate (GHA) score based on observations by parents, teachers, and independent classroom assessors, as well as computerized attention tests. The study demonstrated statistically significant increases in hyperactive behavior (GHA scores) associated with both Mix A and Mix B across the cohorts, showing that these behavioral effects occurred in the general pediatric population and were not restricted to children with pre-existing ADHD. Although EFSA's AFC Panel

evaluated the study and concluded that it provided "limited evidence of a small effect" with substantial uncertainties (including the inability to isolate the effects of individual colorants from the mixtures), the European Parliament adopted a highly precautionary stance. Under Article 24 of Regulation (EC) No 1333/2008, the EU mandated that any food or beverage containing any of these six colors must carry a prominent warning label stating: "May have an adverse effect on activity and attention in children". This regulatory requirement led to a voluntary phase-out of these colorants by manufacturers across the EU, with the prevalence of these dyes in packaged foods plummeting from approximately 3% pre-2008 to consistently below 0.5% post-2010. Conversely, the US FDA has resisted equivalent warning labels, maintaining that the scientific evidence linking these certified color additives to behavioral impairment is insufficient under US law, illustrating a persistent regulatory divide. These and other active chemical safety evaluations are monitored globally.

Chemical Substance	Primary Functional Class	Regulatory / Toxicological Assessment	Active Post-Market Actions (FDA / EFSA)	Recommended Scoring
Azodicarbonamide (ADA) (E927a)	Dough Conditioner / Flour Bleaching Agent	Undergoes thermal decomposition to semicarbazide, a suspected carcinogen.	FDA post-market safety review initiated on August 19, 2025, under Request for Information (RFI) docket FDA-2026-N-4126.	Hazard Tier: Moderate (Orange). Deduct moderate points and restrict maximum product score to $< 50/100$.
3-MCPD & Glycidyl Esters	Processing Contaminants	Formed during high-temperature refining of edible vegetable oils; mutagenic and carcinogenic in rodent models.	FDA active risk management monitoring (initiated July 2023); engagement with manufacturers to mitigate infant formula levels.	Contaminant Penalty. Apply a dynamic deduction to refined vegetable oils and infant formulas lacking certified mitigation.
4-Methylimidazole (4-MEI)	Manufacturing Impurity	Present in Caramel Colors Class III and Class IV; classified as a potential carcinogen.	FDA active scientific assessment of industrial manufacturing controls to minimize levels in soda and dark sauces.	Hazard Tier: Moderate. Apply a selective penalty for Class III and IV caramel colors to incentivize clean-label alternatives.
Short-Chain PFAS (e.g., 6:2 FTOH)	Food Contact Substance	Endocrine disruptor; bioaccumulates; linked to immunological impairment and developmental toxicity.	Phase-out completed in February 2024; all grease-proofing agents containing PFAS are banned from US paperboard packaging.	Packaging Penalty. Apply a severe deduction for products utilizing grease-proofed paperboard manufactured prior to the phase-out.
Bisphenol A (BPA)	Food Contact Substance	Potent endocrine disruptor; mimics estrogen; linked to reproductive and metabolic disorders.	Under active scientific assessment; FDA reviewing a major Food Additive Petition filed on May 2, 2022, to ban packaging uses.	Packaging Penalty. Apply a direct ingredient concern penalty for canned foods utilizing epoxy resins unless certified BPA-free.

Software Architecture for Ingestive Risk Modeling To translate these complex, non-linear toxicological profiles and processing

metrics into a scalable consumer scoring engine, developers must move away from static, relational database tables in favor of an integrated, graph-based metadata architecture. Graph-Based Ingredient Ontologies Because ingredients and additives exhibit complex hierarchical relationships, they are best represented using a Directed Acyclic Graph (DAG) model, similar to the Open Food Facts taxonomy system. In this graph architecture, each ingredient, allergen, nutrient, and reference point is modeled as a distinct node, with parent-child relationships defining properties such as chemical class, processing degree, and dietary compatibility (e.g., vegan, vegetarian, organic). For example, a graph node representing Sunset Yellow FCF (E110) inherits properties from multiple ancestral paths:



By traversing this graph, the scoring engine can dynamically calculate penalties based on ancestral nodes. If a new azo dye is added to the market, the engine immediately inherits the appropriate behavioral and synthetic-processing penalties from the parent node without requiring manual database updates. Furthermore, this ontology must incorporate robust text-processing pipelines. Raw ingredient statements from packaging labels are highly unstandardized. The engine should utilize localized stopword filters (e.g., ignoring connecting prepositions like "contains" or "made with") and recursive synonym mapping to resolve variant terms (e.g., mapping "yellow 6," "Sunset Yellow," "FD&C Yellow No. 6," and "E110" to the same canonical node). Real-Time Integration with EFSA's OpenFoodTox 3.0 API Rather than hard-coding static chemical risk scores, the scoring engine's backend should dynamically query structured toxicological endpoints. In April 2026, EFSA released OpenFoodTox 3.0, which has been fully migrated into the International Uniform Chemical Information Database (IUCLID 6) in compliance with OECD Harmonised Templates (OHTs). By developing an API client to parse the downloadable Excel export repositories or

query the IUCLID dossier archives, scoring engines can automate the retrieval of validated toxicological parameters for over 7,880 distinct substances and 19,452 reference values. The engine's logic should calculate exposure risk by dividing the estimated daily dietary intake of an additive (EDI) by its established Acceptable Daily Intake (ADI) retrieved from OpenFoodTox 3.0:

$$\text{Hazard Quotient (HQ)} = \frac{\text{EDI}}{\text{ADI}}$$

If $\text{HQ} \geq 1.0$, the engine should trigger an automatic hazard flag, applying a dynamic penalty that scales with the magnitude of the exposure. For uncharacterized chemical mixtures or newly introduced food contact substances where empirical data gaps exist, the software can integrate Quantitative Structure-Activity Relationship (QSAR) models, the Threshold of Toxicological Concern (TTC) values, and AMBIT-2.0 prediction software to estimate biological endpoints based on molecular structure, preventing "regrettable substitutions" by manufacturers.

Actionable Frameworks for Scoring Engine Optimization

To deliver a high-performing food scoring engine, developers should implement the following architectural, mathematical, and toxicological recommendations:

- Composite Architectural Design** Construct a three-tiered scoring engine that balances nutritional density (55–60% weight), additive toxicology (30% weight), and physical processing intensity (10–15% weight). Use organic certification solely as a mathematical modifier for pesticide exposure risk, avoiding flat bonuses that can mask poor nutritional quality.
- Non-Linear Penalty Calculations** Replace flat-subtraction additive penalties with non-linear, concentration-dependent risk curves. Implement a hard mathematical cap on overall product scores (e.g., maximum score of 49/100) only for additives linked to direct, non-threshold hazards such as genotoxicity (e.g., Nitrites) or proven neurobehavioral dysfunction (e.g., Southampton Azo Dyes).
- Matrix-Focused Processing Analysis** Incorporate Siga-style processing indicators to identify ultra-processed formulations. Apply progressive penalties for the presence of multiple cosmetic additives (MUPs), as a high density of these compounds indicates a degraded food matrix and an increased risk of overeating.
- Dynamic API Integration** Integrate the engine's backend with open toxicological repositories, specifically EFSA's OpenFoodTox 3.0. Use these standardized, internationally validated chemical hazard endpoints to automate risk assessments, ensuring the engine's ratings remain scientifically objective and up-to-date.